Gameformer method for trajectory prediction into UniTraj Framework

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Gameformer

We propose to implement a Gameformer model adapted in the unitraj framework. This model was first proposed in [1] based only on the WOMD dataset.

- Gameformer uses game theory to predict trajectories based on future interactions between different agents.
- Can out perform state of the art models with minADE6 = 0.9161 on WOMD dataset.

Game theory

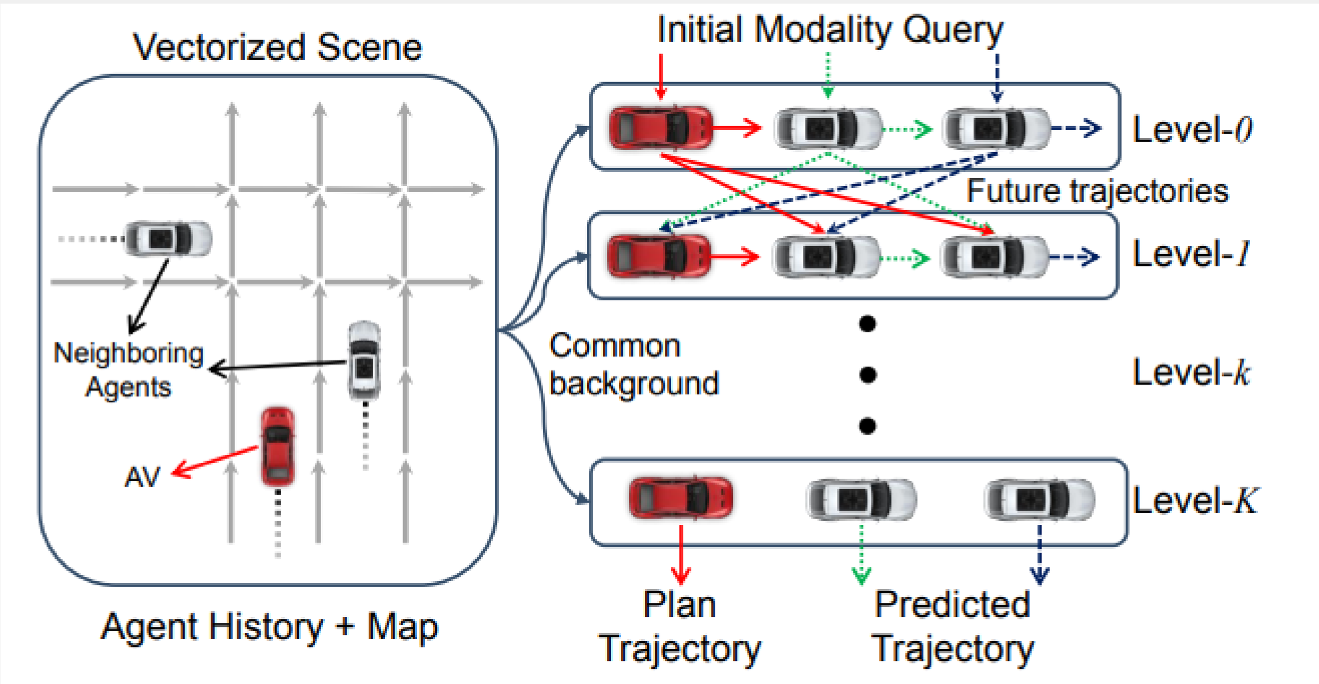


Figure 1. The interactions between agents

- A k-level agent reacts based on the k-1 level agents and scene encoding.
- An agent needs to be aware of the other agents for actions such as merging and lane changing.

Decoder Model

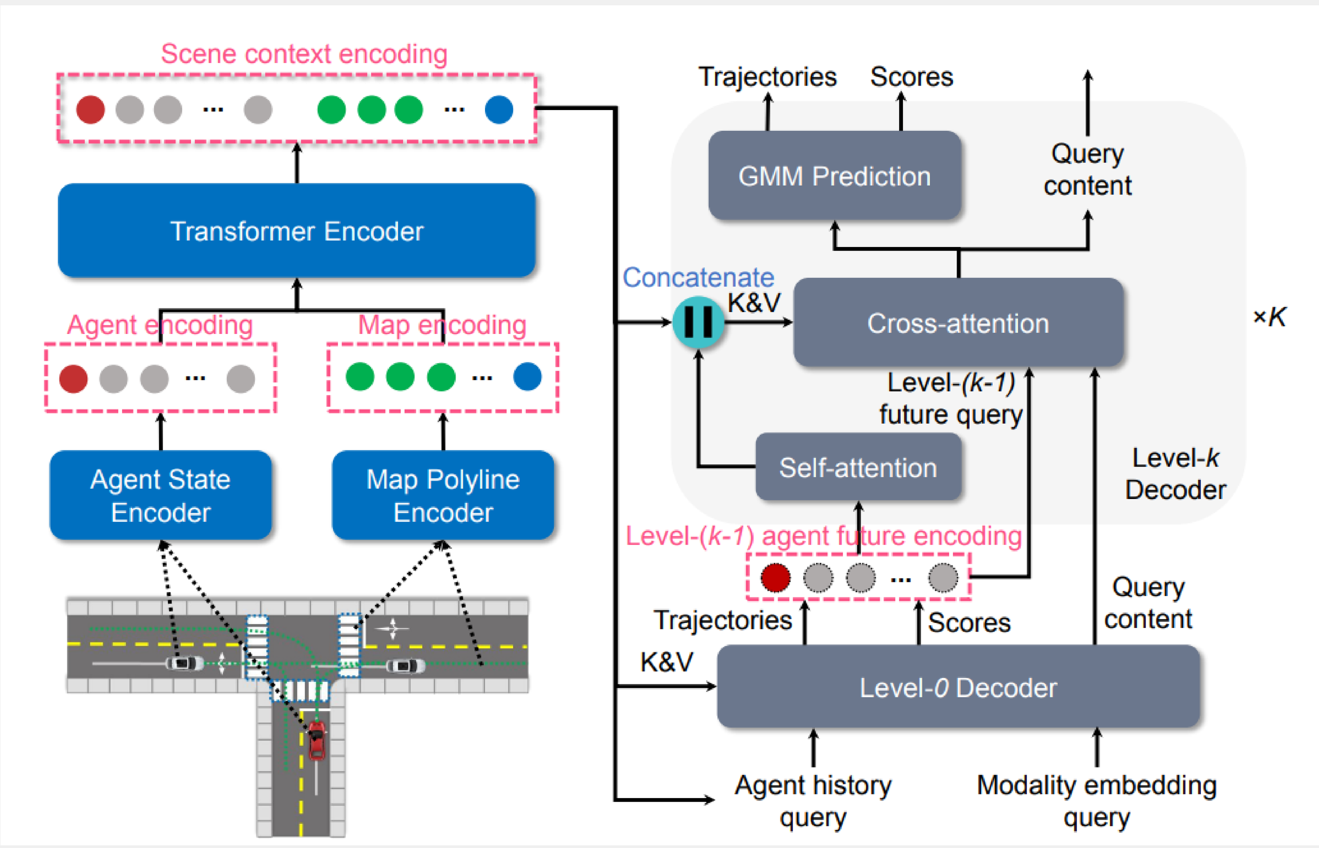


Figure 2. Structure of Gameformer

- The encoder should encode the overall scene for each agent point of view. We choose to keep the ptr encoder in our case adapted to perform on every agents.
- Level 0 decoder predicts firsts trajectories and scores without any interaction process.
- Level k decoder uses future predictions from Level k-1, and the previous encoding as well to perform attention with the predicted future.

Model improvements

Residuals

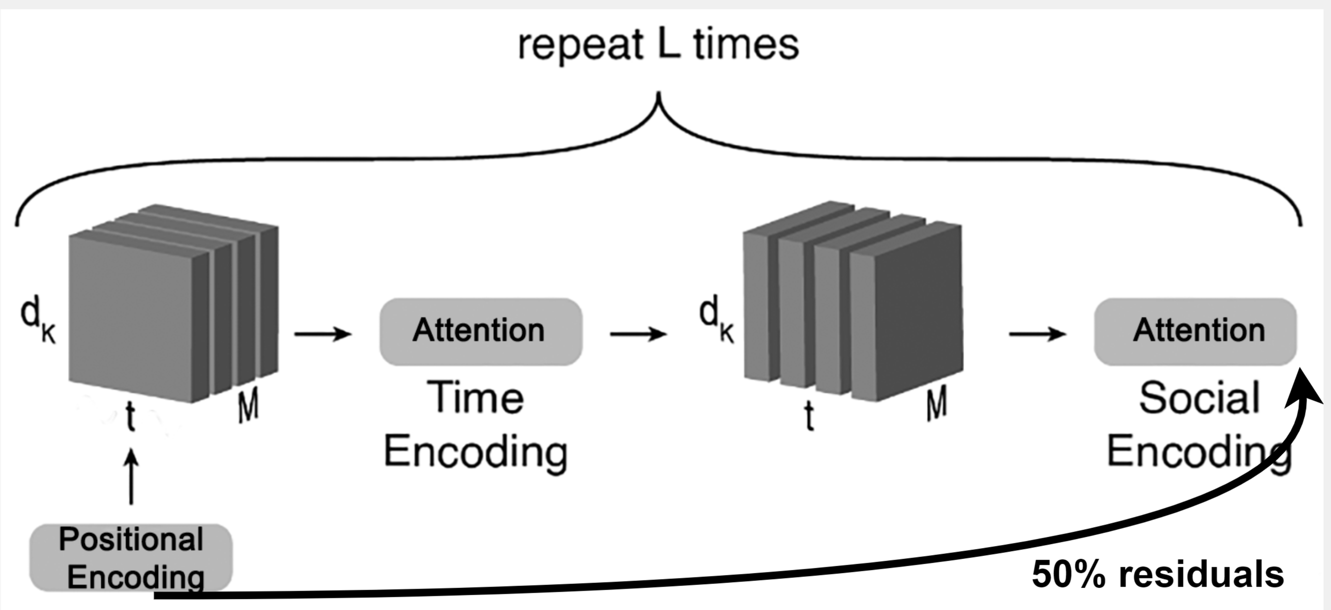


Figure 3. Increasing the complexity

- Augmented model complexity by adding encoder and decoder layers (L_enc = 4 and L_dec = 3).
- To preserve stability, we added 50 % residuals.

Data Augmentation

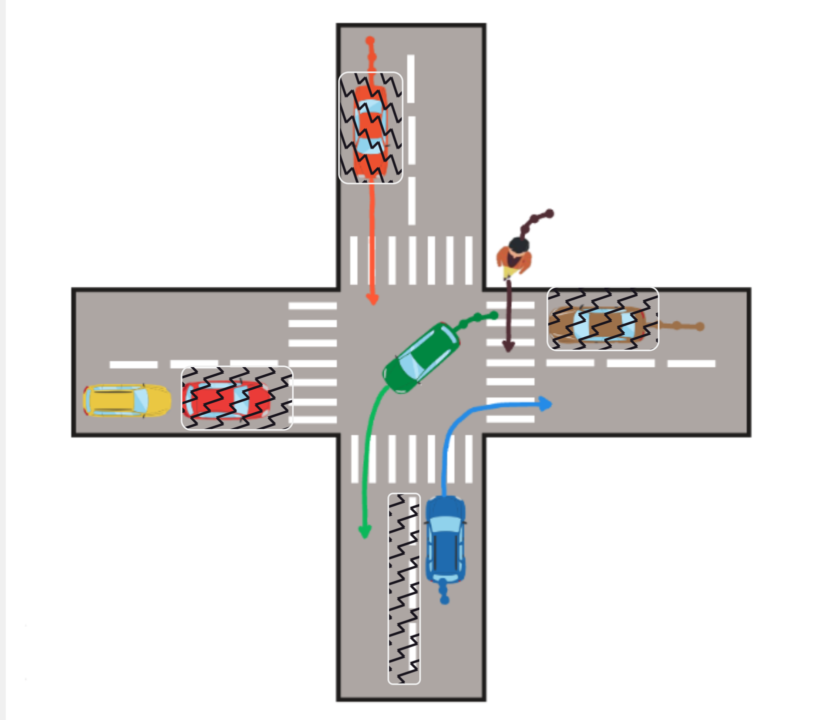
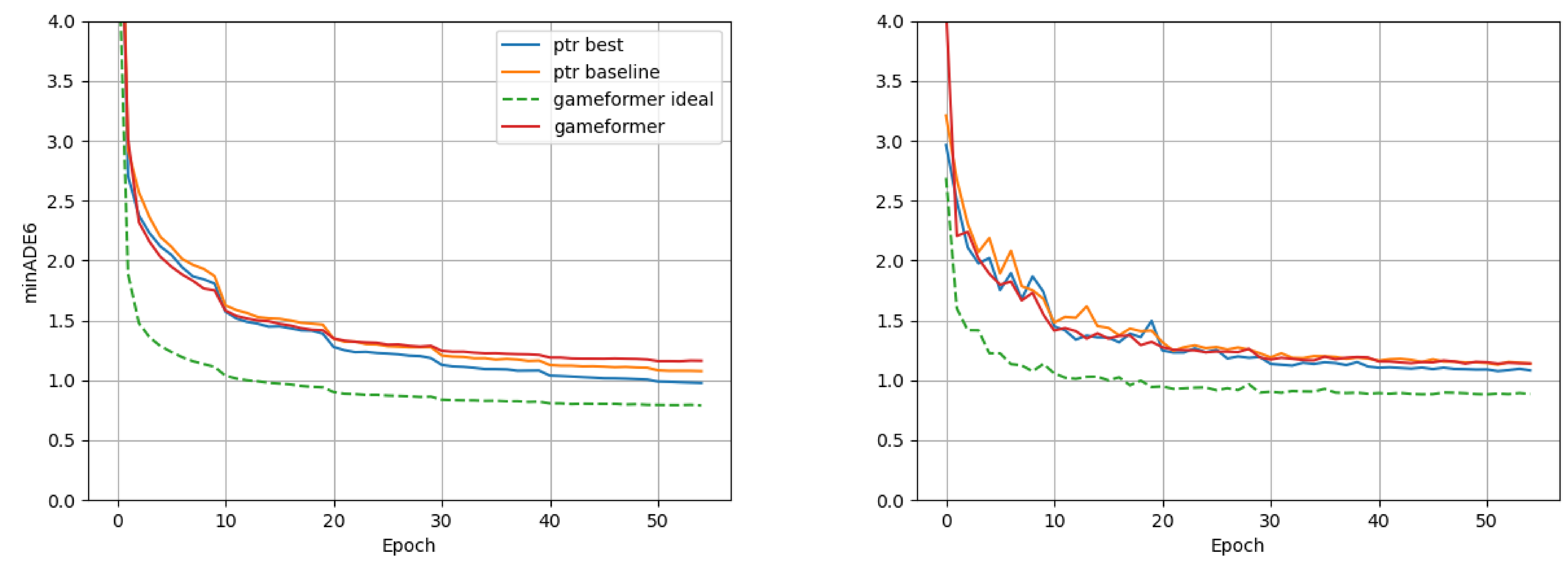


Figure 4. The masks that we applied

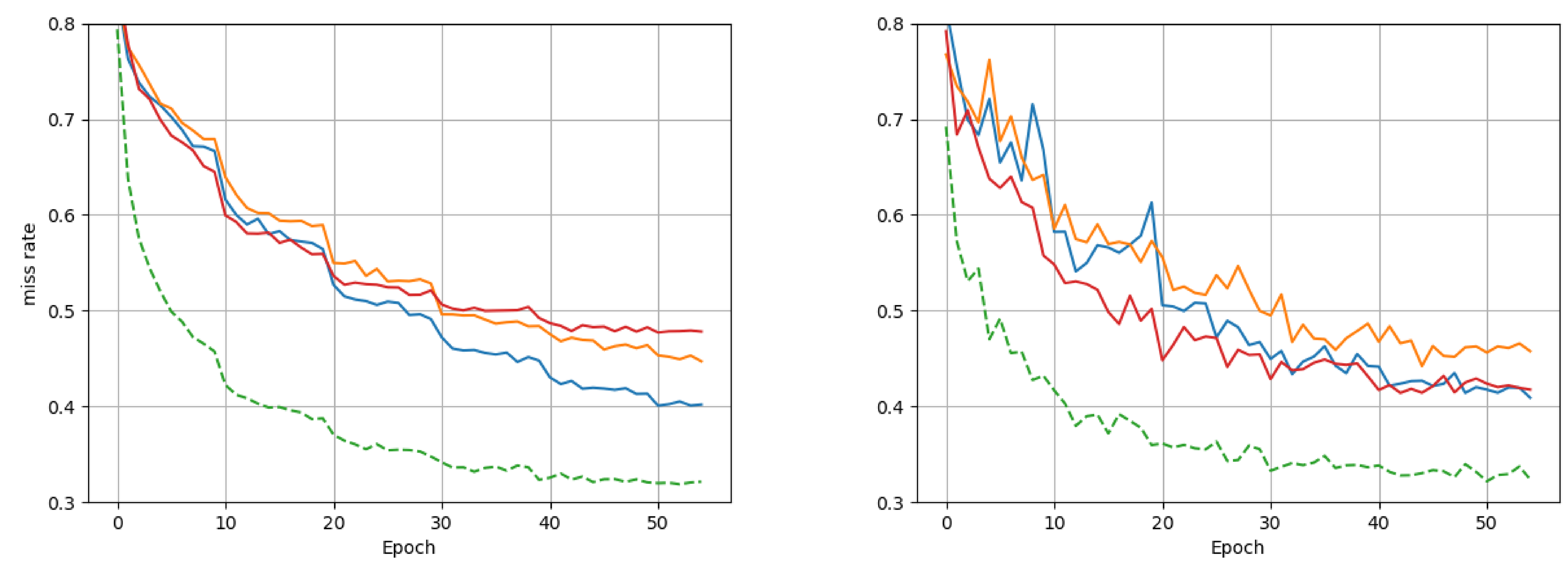
- We masked some agents and some map information to better allow the model to generalize.
- Note : this technique only had a small impact.

Results



(a) training (b) testing

Figure 5. minADE6 training evolution



(a) training (b) testing

Figure 6. miss rate training evolution

	Training set			Validation set			Kaggle hard
	minADE6	minFDE6	miss rate	minADE6	minFDE6	miss rate	minADE6
baseline ptr	0.940	2.055	0.392	1.124	2.508	0.439	...
best ptr	0.920	2.000	0.371	1.087	2.435	0.408	0.989
level 0	...	...	...	1.194	2.788	0.462	...
level 1	...	...	...	1.140	2.600	0.422	...
level 2	...	...	...	1.139	2.603	0.418	1.081
level 3	...	...	...	1.130	2.611	0.424	1.074
level 4	...	...	...	1.163	2.708	0.431	...
level selection	1.164	2.655	0.478	1.141	2.603	0.418	1.081
ideal gameformer	0.790	1.912	0.316	0.894	2.112	0.328	...

Table 1. Metrics overview on training and validation sets

Future Works

- Build a more suitable loss function to better predict best level / mode.
- Use a better initial level predictor and then perform interactions.
- Build upon our improvements by adding other improvements such as handling long tailed distributions.

Conclusion

- Gameformer is competitive with ptr but actually does not improve ptr predictions.
- The potential of this model seems huge but the scores are not well fitted yet.

References

Zhiyu Huang, Haochen Liu, and Chen Lv. Gameformer : Game-theoretic modeling and learning of transformer-based interactive prediction and planning for autonomous driving. 2023.